RAG Retrieval Enhancement and Model Training Samples

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1. Course Content

1. Understand and master the introduction of three concepts: large model hallucination, RAG retrieval enhancement, and model training samples.

> **Tip**
>
> - This lesson is a purely theoretical lesson. If you want to quickly get started and experience the functional gameplay, you can skip this lesson first.

2. What Is Large Model Hallucination? Why Do Large Models Produce Hallucinations?

The main purpose of RAG retrieval enhancement and model training samples is to reduce large model hallucination. Here, let us first understand large model hallucination.

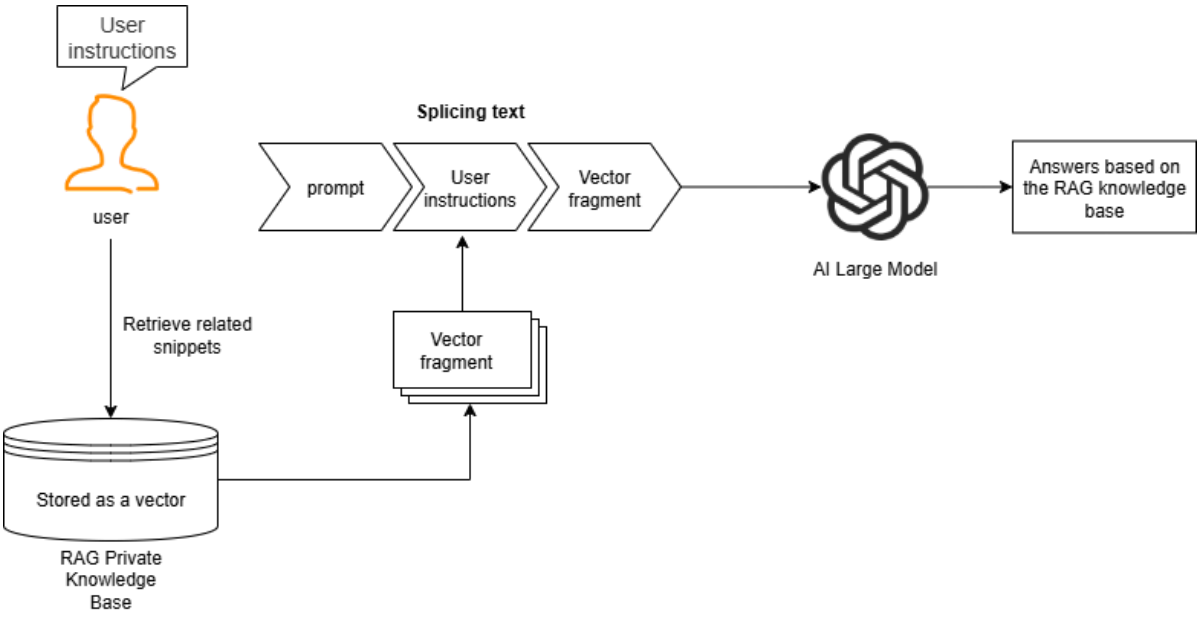
Large model hallucination refers to the model outputting content that does not match real-world facts or logic when understanding the environment, generating instructions, or planning actions, causing the robot to behave incorrectly or causing the user's expected results to deviate significantly.

Why do hallucinations occur? **A large model is essentially a probabilistic prediction system trained on massive text**. In certain specific domains or unknown task scenarios, the training data may lack sufficient samples, causing the model to "fabricate" content to fill in the gaps.

3. RAG Retrieval Enhancement and Training Samples

3.1 RAG Retrieval Enhancement

RAG is an **architecture that combines a retrieval system with a generative model, aiming to solve the limitations of traditional generative models when handling scenarios such as open-domain knowledge question answering and real-time information queries** (such as outdated knowledge, factual hallucinations, and long-text dependencies). Its core logic is: **obtain relevant knowledge through retrieval and integrate it into the Prompt, so that the large model can refer to the corresponding knowledge**, realizing "retrieve first, then generate".



3.2 Training Samples

Training samples are contained in the RAG knowledge base. At the factory, the robot is preset with two knowledge bases: the action function library and the training samples. The training samples contain the training samples from the course case scenarios, providing ideas for the large model's decision-making and planning in specific scenarios. The subsequent [AI Large Model Development - Deploying a Local RAG Knowledge Base] chapter will explain how to configure and extend proprietary knowledge bases and training samples.

4. The Role of RAG Retrieval Enhancement and Training Samples on the Robot

4.1 Reducing Large Model Hallucination

When a large model encounters a domain it has never been exposed to and special user requirements, the large model will give a possible result based on the existing training data, but this result may differ from the result the user expects, or may not meet the requirements of a specific scenario.

4.2 Increasing the Robot's Scenario Generalization Ability

At the factory, the robot is already preset with training samples related to the course cases (explained in detail in the large model configuration course). These training samples provide the large model with reference information in specific scenarios. Users can also add their own training samples to customize their own exclusive robot to adapt to different application domains.